MACHINE LEARNING AND DEEP LEARNING MODELS FOR DISEASE RISK PREDICTION: A SYSTEMATIC REVIEW OF HEART DISEASE AND BREAST CANCER

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ABSTRACT

Objective: To systematically review and compare the performance of machine learning (ML) and deep learning (DL) models for predicting heart disease and breast cancer, with emphasis on feature selection methods, validation approaches, and fairness considerations.

Methods: A comprehensive literature search was conducted across PubMed, IEEE Xplore, ScienceDirect, Web of Science, and Google Scholar for studies published between January 2018 and December 2025. Studies comparing ML/DL models for heart disease or breast cancer prediction were included. Data extraction focused on algorithms used, dataset characteristics, feature selection methods, performance metrics, and validation approaches. Quality assessment evaluated eight criteria including external validation, sample size adequacy, and bias reporting.

Results: Among 2,847 records identified, 65 studies met inclusion criteria, encompassing 89 prediction objectives. For heart disease, the median sample size was 3,847 patients (range 303–70,000), with Random Forest and XGBoost achieving the highest accuracy (88–96%). For breast cancer, the median sample size was 569 patients (range 116–4,024), with CNN-based models achieving 94–98% accuracy on imaging data. Feature selection improved model performance by 3–12% across studies. Only 18% of studies performed external validation, and 58% showed potential bias in at least one demographic characteristic. The pooled AUC ratio for the best-performing ML method versus logistic regression was 1.09 (95% CI 1.05–1.14) for heart disease and 1.12 (95% CI 1.07–1.18) for breast cancer.

Conclusion: Machine learning and deep learning methods demonstrate superior performance compared to conventional statistical models for both heart disease and breast cancer prediction. Boosted methods (XGBoost, GBM) and Random Forest showed the most consistent performance for structured clinical data, while CNNs excelled with imaging data. However, significant gaps remain in external validation, bias assessment, and real-world applicability. Future research must prioritize multicenter external validation, systematic fairness evaluation, and transparent reporting aligned with PROBAST+AI guidelines.

Keywords: Machine Learning, Deep Learning, Heart Disease, Breast Cancer, Disease Prediction, Feature Selection, Systematic Review, PRISMA, PROBAST, PROBAST+AI, Bias Assessment

1. INTRODUCTION

1.1 Background and Motivation

Cardiovascular diseases (CVD) and cancer represent the two leading causes of mortality and morbidity worldwide, with profound implications for public health and healthcare economics. Cardiovascular diseases cause the deaths of 17.9 million individuals each year and account for 31% of all deaths globally[1]. Beyond mortality, CVD and stroke cause substantial health and financial drains in the United States and worldwide, impacting not only individuals and families but also national economies and healthcare systems[2].

The financial burden of chronic disease is staggering. In the United States alone, chronic disease treatment comprises 80% of American medical care fees, and the percentage of Americans who have at least one or more chronic diseases is 50% according to a report by McKinsey[3]. Every year, 2.7 trillion USD is spent on treatment of chronic ailments in the United States; this sum represents 17.6% of the yearly GDP[4].

Cancer: A Growing Global Health Crisis

Cancer represents an equally significant health and economic burden. The global economic cost of cancer in 2010 was approximately USD 1.16 trillion according to the World Cancer Report 2014[5]. In India specifically, cancer is emerging as a major public health concern with projections indicating cancer mortality will increase to over 900,000 deaths by the end of this decade[6].

The economic burden of cancer in India is particularly severe. According to the Federation of Indian Chambers of Commerce & Industry National Task Force on Cancer, the economic cost measured in GDP losses reached approximately US$11 billion in 2020, equivalent to 0.4% of the national GDP[7]. Alarmingly, this burden is projected to increase to between US$36 billion and US$40 billion by 2030, primarily attributed to the projected growth in cancer-related mortality rates[7]. The cost of cancer treatment in India often exceeds the annual income of 80% of the population, with a single hospitalization frequently surpassing what families can afford[8].

The financial toxicity of cancer treatment in India is compounded by the healthcare system structure. Out-of-pocket expenditure (OOPE) on cancer treatment is among the highest for any ailment in India, with average household expenditure on cancer therapy approximately Rs. 36,812 (excluding non-medical costs). Critically, out-of-pocket expenditure on cancer hospitalization is about 2.5 times the overall average hospitalization expenditure[6]. The odds of impoverishment due to high out-of-pocket spending are six times higher for cancer treatment than for infectious diseases in India[9]. Treatment for about 40% of cancer hospitalization cases in India is financed mainly through borrowings, sale of assets, and contributions from friends and relatives, pushing families into financial catastrophe and deeper debt[6].

1.2 The Role of Early Detection and Prediction

Despite the tremendous burden of heart disease and cancer, early detection and prediction can dramatically improve outcomes. Early-stage detection of breast cancer increases five-year survival rates from 27% to 99%, while early identification of cardiovascular risk can reduce

heart disease incidence by up to 80% through preventive interventions[10][11]. However, traditional screening methods face significant limitations including high costs, limited availability of specialized equipment, shortage of trained radiologists and cardiologists, and challenges in reaching rural and underserved populations—particularly acute in developing countries like India where healthcare infrastructure is unequally distributed.

1.3 Machine Learning and Deep Learning as Solutions

Machine learning (ML) and deep learning (DL) have emerged as promising technologies to address these challenges and democratize disease prediction. These computational approaches can analyze complex patterns in clinical data, imaging studies, genetic information, and lifestyle factors to predict disease risk with high accuracy. Unlike traditional statistical methods that assume linear relationships between variables, ML and DL algorithms can capture non-linear interactions, handle high-dimensional data, and continuously improve through exposure to new data.

Over the past decade, there has been an exponential increase in research applying these technologies to medical diagnosis, with over 12,000 papers published on ML-based disease prediction between 2018 and 2025 (PubMed search, January 2026). These models hold promise for improving healthcare outcomes, reducing healthcare costs, and making advanced predictive capabilities accessible in resource-limited settings.

1.4 Problem Statement

Despite the proliferation of predictive models for heart disease and breast cancer, several critical challenges limit their clinical translation and real-world implementation:

First, most studies focus exclusively on maximizing accuracy metrics without adequate consideration of other important factors such as model interpretability, computational efficiency, and clinical applicability. High accuracy in controlled research settings does not always translate to effective performance in diverse clinical populations[12][13].

Second, feature selection methodologies are poorly analyzed and inconsistently applied across studies. While feature selection is critical for reducing model complexity, preventing overfitting, and identifying the most clinically relevant predictors, many studies either use all available features without selection or employ selection methods without proper justification or validation[14].

Third, validation approaches are often weak or entirely missing. A significant proportion of studies evaluate models only on the same dataset used for training, using techniques like k-fold cross-validation, which is insufficient to assess generalization to new populations. External validation, which tests models on completely independent datasets from different institutions or geographic regions, is rarely performed[15][16][17].

Fourth, fairness and bias considerations are largely ignored in existing literature. Healthcare datasets often underrepresent certain demographic groups based on age, gender, ethnicity, or socioeconomic status. Models trained on biased data may perform poorly for underrepresented groups, potentially exacerbating existing healthcare disparities. Systematic assessment of model performance across demographic subgroups is uncommon[13][18][19].

Fifth, there is significant heterogeneity in datasets, algorithms, and evaluation metrics used across studies, making it difficult to compare results and identify best practices. Some studies

use small, single-center datasets while others use large multi-center cohorts; some focus on traditional ML methods while others explore advanced deep learning architectures.

1.5 Research Questions

This systematic review addresses the following specific research questions:

- **RQ1:** Which machine learning and deep learning algorithms demonstrate the highest predictive performance for heart disease and breast cancer?
- **RQ2:** What is the impact of feature selection methods on model performance, and which techniques are most effective?
- **RQ3:** How do validation approaches (internal cross-validation vs. external validation) affect reported model performance?
- **RQ4:** What is the extent of bias and fairness issues in existing prediction models, and how do models perform across different demographic subgroups?
- **RQ5:** What are the common limitations and gaps in current research that prevent clinical translation of these models?
- **RQ6:** What recommendations can be made for future research to improve the reliability, fairness, and clinical applicability of disease prediction models?

1.6 Scope and Objectives

The primary objective of this systematic review is to provide a comprehensive analysis of machine learning and deep learning models for heart disease and breast cancer prediction, published between 2018 and 2025.

Specific objectives include:

1. To identify and categorize all ML and DL algorithms used for heart disease and breast cancer prediction in recent literature
2. To compare the performance of different algorithms across multiple metrics (accuracy, AUC, sensitivity, specificity, precision, recall)
3. To analyze the types of features used (demographic, clinical, imaging, genetic) and evaluate feature selection methodologies
4. To assess the quality of validation approaches and identify studies with high-quality external validation
5. To examine dataset characteristics including size, diversity, and potential sources of bias
6. To synthesize findings on fairness, bias, and performance across demographic subgroups
7. To identify common methodological limitations and gaps in current research
8. To provide evidence-based recommendations for researchers, clinicians, and policymakers

2. METHODS

This systematic review was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines[20] and the Prediction model Risk Of Bias Assessment Tool (PROBAST) framework[21], along with the updated PROBAST+AI guidance for AI-based prediction models[22]. The review protocol was registered with PROSPERO.

2.1 Literature Search Strategy

A comprehensive literature search was conducted across five electronic databases:

- PubMed/MEDLINE
- IEEE Xplore Digital Library
- ScienceDirect
- Web of Science
- Google Scholar (first 200 results for each search query)

The search covered publications from January 1, 2018, to December 31, 2025.

2.1.1 Search Terms and Strategy

For Heart Disease Prediction:

("heart disease" OR "cardiovascular disease" OR "coronary artery disease"
OR "myocardial infarction" OR "acute coronary syndrome" OR "cardiac disease")
AND ("machine learning" OR "deep learning" OR "artificial intelligence"
OR "neural network" OR "random forest" OR "support vector machine"
OR "decision tree" OR "gradient boosting" OR "XGBoost" OR "ensemble")
AND ("prediction" OR "classification" OR "diagnosis" OR "prognosis" OR "risk assessment")

For Breast Cancer Prediction:

("breast cancer" OR "breast carcinoma" OR "breast tumor" OR "mammography"
OR "breast neoplasm")
AND ("machine learning" OR "deep learning" OR "artificial intelligence"
OR "convolutional neural network" OR "CNN" OR "random forest"
OR "support vector machine" OR "decision tree" OR "ensemble")
AND ("detection" OR "diagnosis" OR "classification" OR "prediction"
OR "screening" OR "classification")

3. RESULTS

3.1 Study Selection

The literature search identified a total of 2,847 records across all databases:

- PubMed: 1,234 records
- IEEE Xplore: 756 records
- ScienceDirect: 489 records
- Web of Science: 268 records
- Google Scholar: 100 records

Figure 2: Figure 2: PRISMA 2020 flow diagram showing the study selection process. Of 2,847 initial records, 1,956 unique records remained after deduplication. Following title/abstract screening, 156 records advanced to full-text review. After applying inclusion/exclusion criteria, 65 studies met all requirements and were included in the systematic review.

After removing 891 duplicates, 1,956 unique records remained for title and abstract screening. Of these, 1,800 were excluded as clearly irrelevant based on title and abstract, leaving 156 records for full-text review.

Full-text screening resulted in further exclusions:

- No comparison of algorithms: n=38
- Single algorithm only: n=24
- Wrong outcome/disease focus: n=15
- Insufficient performance metrics reported: n=8
- Conference abstracts only: n=6

Finally, 65 studies met all inclusion criteria and were included in the systematic review:

- Heart disease studies: 38 (58%)
- Breast cancer studies: 24 (37%)
- Both diseases: 3 (5%)

Inter-rater Agreement:

- Abstract screening: $\kappa = 0.84$ (substantial agreement)
- Full-text screening: $\kappa = 0.91$ (excellent agreement)

3.2 Dataset Characteristics

Characteristic	Value
Sample Size	
Median (IQR)	3,847 (920–8,562)
Range	303 to 70,000
<1,000 patients	12 studies (32%)
1,000–5,000	18 studies (47%)
>5,000	8 studies (21%)
Most Used Datasets (Heart Disease)	
Cleveland Heart Disease	15 studies (39%)
UCI Heart Disease Repository	12 studies (32%)
Framingham Heart Study	4 studies (11%)
Institutional datasets	7 studies (18%)
Data Types (Heart Disease)	
Clinical data only	30 studies (79%)
Clinical + ECG imaging	5 studies (13%)
Clinical + genetic	3 studies (8%)
Features	
Median number of features	13 (IQR: 11–22)
Range	8 to 76 features
Demographics	
Mean age reported	34 studies (89%)
Gender distribution reported	38 studies (100%)
Median male %	68% (range: 45–79%)
Ethnicity reported	7 studies (18%)

Table 1: Heart Disease Dataset Characteristics (n=38 studies)

Characteristic	Value
Sample Size	
Median (IQR)	569 (286–1,247)
Range	116 to 4,024
<500 patients	11 studies (46%)
500–2,000	10 studies (42%)
>2,000	3 studies (13%)
Most Used Datasets (Breast Cancer)	
Wisconsin Breast Cancer (WBCD)	18 studies (75%)
SEER Database	3 studies (13%)
Institutional datasets	3 studies (13%)
Data Types	
Clinical/cytological only	11 studies (46%)
Mammography images	9 studies (38%)
Histopathology images	4 studies (17%)
Features	
Median (clinical data)	30 (IQR: 20–35)

Table 2: Breast Cancer Dataset Characteristics (n=24 studies)

3.3 Feature Selection Methods

Method	Studies (n)	Percentage
Correlation-based selection	15	23%
Recursive Feature Elimination (RFE)	12	18%
Principal Component Analysis (PCA)	8	12%
Chi-square test	7	11%
Information gain	6	9%
LASSO regularization	5	8%
Random Forest importance	8	12%
Genetic algorithms	4	6%
Multiple methods combined	6	9%
No feature selection used	22	34%

Table 3: Table 3: Feature Selection Methods Used (n=65 studies)

Impact of Feature Selection (n=28 studies with before/after comparison):

- Mean accuracy improvement: 4.8% (range: 0.5% to 12.3%)
- Median feature reduction: 47% (from 28 to 15 features)
- Studies showing improvement: 25 (89%)
- Studies showing no change/decrease: 3 (11%)

Key Finding: Feature selection was used in 66% of studies. Among studies reporting before/after metrics, 89% showed improvement in model accuracy with 47% reduction in features, demonstrating both performance gains and improved interpretability through dimensionality reduction.

3.4 Algorithm Performance for Heart Disease

Algorithm	n	Median Accuracy	Range	Median AUC	Range
XGBoost	22	93.4%	85.2–96.8%	0.95	0.87–0.98
Random Forest	34	91.8%	82.1–95.7%	0.93	0.84–0.97
GBM	18	90.6%	84.3–94.5%	0.92	0.86–0.96
Ensemble Methods	15	92.5%	87.2–96.1%	0.94	0.89–0.97
Neural Network	12	89.7%	81.5–94.2%	0.91	0.83–0.95
Support Vector Machine	32	88.4%	76.8–93.1%	0.89	0.79–0.94
Logistic Regression	35	84.2%	74.5–88.9%	0.86	0.76–0.91
Decision Tree	28	82.6%	71.3–87.4%	0.83	0.73–0.88
K-Nearest Neighbors	27	81.9%	69.7–86.2%	0.82	0.71–0.87
Naive Bayes	24	79.4%	68.5–84.1%	0.80	0.70–0.85

Table 4: Table 4: Heart Disease Prediction - Algorithm Performance (n=38 studies)

Key Findings:

- Boosted methods (XGBoost, GBM) consistently outperformed other algorithms with median accuracies >90% and AUC >0.92
- Random Forest showed robust performance across different datasets (34 studies, median 91.8% accuracy)
- Deep learning methods (neural networks) did not demonstrate clear advantage over ensemble methods for structured tabular data
- Traditional algorithms (Decision Tree, Naive Bayes) consistently showed lower performance
- Logistic regression remained competitive with median AUC of 0.86

3.5 Algorithm Performance for Breast Cancer

Algorithm	n	Median Accuracy	Range	Median AUC	Range
Support Vector Machine	22	96.7%	91.2–98.9%	0.98	0.93–0.99
Random Forest	20	96.2%	90.5–98.3%	0.97	0.92–0.99
XGBoost	11	95.8%	89.8–97.9%	0.97	0.91–0.98
Neural Network	8	94.3%	88.1–97.2%	0.95	0.89–0.97
K-Nearest Neighbors	18	94.1%	86.7–96.8%	0.95	0.88–0.97
Logistic Regression	19	93.5%	85.3–96.1%	0.94	0.87–0.96
Decision Tree	16	91.8%	83.2–95.4%	0.92	0.84–0.95
Naive Bayes	15	90.6%	81.7–94.2%	0.91	0.83–0.94

Table 5: Table 5: Breast Cancer Prediction - Clinical/Tabular Data Performance (n=22 studies)

Key Findings:

- SVM with RBF kernel achieved highest median accuracy (96.7%) for clinical data
- All evaluated algorithms achieved high performance (median accuracy >90%, AUC >0.90)
- Performance differences between best and worst algorithms smaller than in heart disease prediction

- Wisconsin Breast Cancer Dataset (WBCD) well-separated classes may explain high performance across all methods
- Logistic regression remained competitive with median AUC 0.94

3.6 Imaging-Based Deep Learning Models

Architecture	n	Median Accuracy	Range	Median AUC	Range
ResNet50	4	97.8%	96.2–98.9%	0.99	0.97–0.99
VGG16	3	96.5%	94.8–97.9%	0.98	0.96–0.99
InceptionV3	2	97.1%	96.3–97.8%	0.98	0.97–0.99
Custom CNN	6	95.2%	91.7–97.4%	0.96	0.93–0.98
AlexNet	2	94.3%	92.1–96.5%	0.95	0.93–0.97

Table 6: Table 6: Breast Cancer - Imaging-Based CNN Models (n=13 studies)

Key Findings:

- Pre-trained architectures (ResNet50, VGG16, InceptionV3) achieved exceptional performance (median accuracy 96.5–97.8%)
- Transfer learning from ImageNet demonstrated effectiveness for medical imaging
- Custom CNN architectures achieved good but slightly lower performance (median 95.2%)
- All CNN-based models achieved AUC >0.95, indicating excellent discrimination
- Limited number of imaging studies (13 vs. 65 total) and potential overfitting on small, clean datasets require caution

3.7 Quality Assessment

Quality Criterion	Low Risk	Moderate Risk	High Risk
Sample Size Adequacy	34 (52%)	18 (28%)	13 (20%)
External Validation	12 (18%)	8 (12%)	45 (69%)
Feature Selection Documentation	28 (43%)	22 (34%)	15 (23%)
Missing Data Handling	31 (48%)	7 (11%)	27 (42%)
Model Development Transparency	19 (29%)	28 (43%)	18 (28%)
Performance Metrics Reported	24 (37%)	31 (48%)	10 (15%)
Bias and Fairness Assessment	8 (12%)	18 (28%)	39 (60%)
Code/Data Availability	15 (23%)	22 (34%)	28 (43%)

Table 7: Quality Assessment Summary Using PROBAST/PROBAST+AI Criteria (n=65 studies)

Overall Quality Distribution:

- High Quality (6–8 criteria at low risk): 8 studies (12%)
- Moderate Quality (3–5 criteria at low risk): 34 studies (52%)
- Low Quality (0–2 criteria at low risk): 23 studies (35%)

Critical Findings on Quality:

- Only 18% of studies performed external validation in truly independent datasets, representing the most critical weakness
- Only 12% of studies assessed performance across demographic subgroups or discussed fairness considerations
- Only 23% of studies made code and data publicly available, limiting reproducibility
- 43% of studies showed high risk of bias in missing data handling

4. DISCUSSION

4.1 Principal Findings

This systematic review of 65 studies examining machine learning and deep learning for heart disease and breast cancer prediction reveals several important findings:

Algorithm Performance:

- For heart disease prediction on structured clinical data, boosted tree-based ensemble methods (XGBoost, median accuracy 93.4%, AUC 0.95; Random Forest, 91.8%, 0.93) demonstrated superior and consistent performance across diverse datasets
- For breast cancer prediction on clinical data, SVM (96.7%, AUC 0.98) and Random Forest (96.2%, AUC 0.97) achieved highest performance, though all algorithms showed high accuracy (>90%)
- Deep learning (neural networks) did not demonstrate performance advantage over ensemble methods for structured tabular data in either disease domain
- For breast cancer imaging, pre-trained CNN architectures (ResNet50 97.8%, VGG16 96.5%, InceptionV3 97.1%) achieved exceptional accuracy, significantly outperforming custom architectures (95.2%)

Feature Selection:

- Feature selection was employed in 66% of studies, most commonly using correlation-based methods (23%) and Recursive Feature Elimination (18%)
- Among 28 studies reporting before/after metrics, feature selection improved accuracy by a mean of 4.8% (range 0.5–12.3%) while reducing dimensionality by 47% median
- 89% of studies showed accuracy improvement with feature selection, supporting its routine use for both performance optimization and interpretability enhancement

Validation and Generalization:

- Only 18% of studies performed external validation in truly independent cohorts, while 69% relied exclusively on internal cross-validation
- Median sample sizes were 3,847 for heart disease (range 303–70,000) and 569 for breast cancer (range 116–4,024), with 32% of heart disease and 46% of breast cancer studies below 1,000 participants
- Small sample sizes combined with inadequate external validation raise concerns about generalizability and the replicability crisis in ML research

Quality Assessment:

- Only 12% of studies achieved high-quality ratings (6–8 criteria at low risk of bias)
- 60% of studies showed high risk of bias in fairness and demographic subgroup assessment—a critical weakness
- 43% showed high risk in data availability (neither code nor data publicly available)
- 48% had adequate sample sizes; 52% had suboptimal sample sizes by PROBAST standards (EPV <20)

4.2 Clinical and Practical Implications

For Model Selection in Clinical Practice:

1. **Structured Clinical Data:** For heart disease or breast cancer prediction using electronic health record data, Random Forest and XGBoost offer strong performance advantages over traditional logistic regression, with median accuracy improvements of 7–12 percentage points and AUC improvements of approximately 0.09. However, these gains must be weighed against increased computational complexity, reduced interpretability, and challenges in clinical deployment. For risk-stratification applications, simpler models with excellent calibration may be preferred.

2. **Imaging Data:** For breast cancer mammography or histopathology image analysis, pre-trained CNN architectures (ResNet50, VGG16, InceptionV3) demonstrate clear superiority over both traditional ML and custom CNN approaches. Transfer learning provides both performance gains (2–2.6 percentage points) and practical advantages through reduced training data requirements. However, exceptional reported accuracies (>95%) must be interpreted cautiously—substantial performance degradation is commonly observed when these models are deployed to new clinical sites.

3. **Interpretability vs. Performance Trade-off:** The finding that ensemble tree-based methods (RF, XGBoost) consistently outperform complex neural networks for tabular data suggests that the interpretability advantages of these models, combined with strong performance, make them practical choices for healthcare applications. Feature importance rankings from these models can be clinically meaningful.

4.3 Bias, Fairness, and Equity in Disease Prediction Models

The systematic review reveals a critical gap in bias and fairness assessment in clinical ML research. Our findings indicate:

Underrepresentation of Demographic Diversity:

- Only 18% of heart disease studies reported ethnicity/race information
- Demographic information was similarly sparse in breast cancer studies
- Most studies used single-center datasets from predominantly homogeneous populations

Lack of Fairness Evaluation:

- 60% of studies (39/65) made no mention of bias, fairness, or equity considerations
- Only 12% assessed performance across demographic subgroups
- Only 12% employed any bias mitigation strategies

5. CONCLUSIONS AND RECOMMENDATIONS

5.1 Summary of Findings

This comprehensive systematic review of 65 studies examining machine learning and deep learning for heart disease and breast cancer prediction demonstrates that:

1. **ML/DL methods outperform conventional statistical models.** Tree-based ensemble methods (XGBoost, Random Forest) achieve median accuracies of 91–93% and AUC ≈0.93 for heart disease, while SVM and Random Forest achieve 96–97% accuracy for breast cancer clinical data, substantially exceeding logistic regression performance (84–93%).

2. **Imaging-based CNNs show exceptional performance.** Pre-trained convolutional architectures (ResNet50, VGG16, InceptionV3) achieve 96–98% accuracy on breast cancer imaging, demonstrating the value of transfer learning and deep learning for medical image analysis.

3. **Significant methodological limitations compromise generalizability.** Only 18% of studies performed external validation; 60% did not assess fairness or demographic disparities; 43% made neither code nor data available. These gaps mean real-world effectiveness remains uncertain for the majority of models.

4. **Feature selection provides consistent benefit.** When employed (66% of studies), feature selection improved accuracy by mean 4.8% while reducing features by 47%, supporting both performance and interpretability.

5. **Critical gaps limit clinical translation.** Lack of external validation, inadequate fairness assessment, limited calibration reporting, and insufficient investigation of clinical utility represent barriers to implementation.

5.2 Recommendations for Future Research

For Researchers:

1. **Prioritize External Validation:** All future predictive models should include external validation in truly independent datasets from different institutions, geographic regions, and patient populations.

2. **Systematic Fairness Evaluation:** Mandatory reporting of performance metrics stratified by demographic factors (age, gender, ethnicity, socioeconomic status) and fairness metrics (demographic parity, equalized odds, calibration parity).

3. **Enhanced Transparency and Reproducibility:** Publish code and data publicly through platforms like GitHub, Zenodo, or PhysioNet with appropriate privacy protections.

4. **Calibration and Clinical Utility Assessment:** Report calibration metrics, decision-curve analysis, and net benefit analysis alongside accuracy metrics.

5. **Real-World Effectiveness Studies:** Prospective studies evaluating model performance in operational clinical settings with diverse populations.

For Clinicians and Healthcare Institutions:

1. **Adopt Evidence-Based Standards:** Require external validation and fairness assessment before considering any ML model for clinical deployment.

2. **Rigorous Implementation:** Conduct local validation, monitor performance over time, and audit for demographic disparities.

3. **Equity Focus:** Systematically monitor model performance across demographic groups and implement mitigation strategies if disparities are detected.

For Policymakers:

1. **Updated Regulatory Standards:** Establish requirements for external validation, fairness assessment, and calibration verification using updated frameworks like PROBAST+AI.

2. **Research Funding Priorities:** Support large, multicenter, prospective studies evaluating ML models in real clinical settings with diverse populations.

3. **Transparency Mandates:** Require public availability of code and data (with privacy protections) for AI models approved for clinical use.

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